

# AHMED ELMETWALLY

Electrical Engineering Graduate, MEng | EIT Application Pending (BC)

Windsor, ON (Able to Relocate) | (226) 340-0506 | [ahmedoelmetwally@gmail.com](mailto:ahmedoelmetwally@gmail.com) | [linkedin.com/in/ahmedoelmetwally](https://www.linkedin.com/in/ahmedoelmetwally)

## SUMMARY OF QUALIFICATIONS

- Built Ontario electricity forecasting and billing models in Excel and Power Query, achieving \$9.89/MWh forecast MAE.
- Secured about US\$120K in savings and cut downtime by up to 66% via a 103,000-device circular economy initiative.
- Produced Revit MEP sheets and AutoCAD Electrical schematics, panel layouts, wiring diagrams, and equipment records.
- Performed power flow, short-circuit, and contingency analysis on a modified IEEE 14-bus system using PSAT and MATLAB.
- Technical Skills:** AutoCAD Electrical, Revit MEP, DIALux, PVsyst, ETAP, PSAT, MATLAB/Simulink, Proteus, Power BI, think-cell, Excel, Power Query, SharePoint, Microsoft Project, and Microsoft Office Suite.

## EDUCATION

### Master of Engineering, Electrical and Computer Engineering Co-operative Education

Jan 2025 - Apr 2026

University of Windsor, Windsor, ON

- Average: 93%, Master of Engineering Honour Roll.
- Coursework: Project Management, Power System Analysis, Energy Storage Systems, and EV Power Conversion.

### Bachelor of Science, Electrical Energy Engineering (ABET-accredited)

Sep 2018 - Jul 2023

Arab Academy for Science and Technology, Cairo, Egypt

- Average: 95%, Excellent with Honours, ranked among the top three students.

## WORK EXPERIENCE

### Battery Materials and Energy Market Research Co-op

Sep 2025 - Dec 2025

PowerCo Canada Inc. (Volkswagen Group), St. Thomas, ON

- Built an Ontario electricity forecast in Excel and Power Query with \$9.89/MWh MAE for cost planning.
- Developed an IESO and Hydro One billing model to forecast facility costs and support invoice validation.
- Prepared quarterly reports on Ontario electricity, policy, solar, battery materials, and mining for senior leadership.
- Evaluated 14 suppliers, 11 PPA/VPPA case studies, and C\$50M+ in incentives for rooftop solar decisions.
- Delivered a Microsoft Copilot workshop to about 40 employees and managers on procurement use cases.

### Technical Vendor Management Engineer

Nov 2023 - Dec 2024

Telecom Egypt, Giza, Egypt

- Coordinated suppliers and internal teams to recycle and refurbish 103,000 telecommunications devices.
- Secured about US\$120K in savings by comparing proposals, negotiating terms, and revising the business plan.
- Reduced downtime by up to 66% across 53 teams through SOPs, RACI matrices, and clear ownership.
- Resolved 84% of 433 vendor issues through technical reviews, action plans, and cross-team follow-up.

### Electrical Maintenance Engineer Intern

Aug 2022 - Sep 2022

Khalda Petroleum Company (Apache), Matrouh, Egypt

- Supported on-site diagnostics and MV maintenance for transformers, ESP motors, and VSDs under HSE rules.

### Renewable Energy Engineer Intern

Jul 2021 - Sep 2021

Smart Solar Company, Cairo, Egypt

- Supported installation and commissioning of a 500-kW rooftop PV system through cabling, inverter setup, and testing.

## SELECTED ENGINEERING PROJECTS

### Power System Analysis and Control, Modified IEEE 14-Bus System

Jan 2026 - Mar 2026

University of Windsor, Windsor, ON

- Performed power flow, short-circuit, and voltage stability analysis on a 14-bus system using PSAT and MATLAB.
- Assessed fault levels, reactive power limits, and N-1 contingency impacts to evaluate system reliability.

### Battery Swapping Station for Electric Scooters

Sep 2022 - Jul 2023

Arab Academy for Science and Technology, Cairo, Egypt

- Led a team to build a prototype integrating power distribution, safety interlocks, sensors, and IoT monitoring.
- Prepared drawings and sized protection using AutoCAD and ETAP, then validated PCB circuits in Proteus.

### Revit MEP Electrical Parking Lot Project

Jun 2026

Revit MEP Electrical Masterclass

- Built a coordinated Revit MEP model covering lighting, fire alarm, receptacles, cable trays, and conduits.
- Coordinated device placement to resolve layout conflicts and produced 12 sheets with schedules and tags.

### AutoCAD Electrical: Motor Control Schematic and Panel Drawing

May 2026

AutoCAD Electrical 2026

- Created motor control schematics, panel layouts, wiring diagrams, BOMs, and PLC I/O documentation.

## CERTIFICATIONS & TRAINING

- Revit MEP Electrical Masterclass - From Beginner to Advanced, Udemy Jun 2026
- AutoCAD Electrical 2026: Tutorials, Exercises, and Projects, Udemy May 2026
- Generative AI for Everyone, DeepLearning.AI Nov 2025
- Electrical Distribution Course (Phase 1 & 2), Schneider Electric & EES Sep 2023